

Report - Runtime Threat Detection with Falco

Starting Colima and Minikube for Local Cluster:

```
~/Documents/Learning/falco -- zsh          -- zsh • kubect1.35.1          -- zsh          +
meshir@NIS-PK564 ~ % colima start --cpus 2 --memory 10
INFO[0002] starting colima
INFO[0002] runtime: docker
INFO[0003] starting ...
INFO[0004] provisioning ...
INFO[0031] starting ...
INFO[0035] done
meshir@NIS-PK564 ~ % minikube start
minikube v1.38.1 on Darwin 13.7.8
🔥 minikube 1.39.0 is available! Download it: https://github.com/kubernetes/minikube/releases/tag/v1.39.0
👉 To disable this notice, run: 'minikube config set WantUpdateNotification false'

👉 Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
📶 Pulling base image v0.0.50 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔍 Verifying Kubernetes components...
   • Using image gcr.io/k8s-minikube/storage-provisioner:v5
   Enabled addons: storage-provisioner, default-storageclass
🎉 Done! kubect1 is now configured to use "minikube" cluster and "default" namespace by default
meshir@NIS-PK564 ~ % clear

meshir@NIS-PK564 ~ % helm repo add falcosecurity https://falcosecurity.github.io/charts
"falcosecurity" has been added to your repositories
meshir@NIS-PK564 ~ % helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "sms-app" chart repository
```

Adding falcosecurity hem repository & installing falco in Minikube:

```
~/Documents/Learning/falco -- zsh -- zsh • kubectll1.35.1 ... -- zsh +
mashir@NIS-PK564 ~ % helm repo add falcosecurity https://falcosecurity.github.io/charts
"falcosecurity" has been added to your repositories
mashir@NIS-PK564 ~ % helm repo update
Hang tight while we grab the latest from your chart repositories...
..Successfully got an update from the "an-up" chart repository
..Successfully got an update from the "falcosecurity" chart repository
..Successfully got an update from the "jenkins" chart repository
..Successfully got an update from the "bitnami" chart repository
..Successfully got an update from the "prometheus-community" chart repository
Update Complete. Happy Helming!
mashir@NIS-PK564 ~ % helm install --replace falco --namespace falco --create-namespace --set tty=true falcosecurity/falco
NAME: falco
LAST DEPLOYED: Mon Sep 21 12:56:29 2026
NAMESPACE: falco
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
TEST SUITE: None
NOTES:
Falco agents are spinning up on each node in your cluster. After a few
seconds, they are going to start monitoring your containers looking for
security issues.

TIP:
You can easily forward Falco events to Slack, Kafka, AWS Lambda and more with falcosidekick.
Full list of outputs: https://github.com/falcosecurity/charts/tree/master/charts/falcosidekick.
You can enable its deployment with "--set falcosidekick.enabled=true" or in your values.yaml.
See: https://github.com/falcosecurity/charts/blob/master/charts/falcosidekick/values.yaml for configuration values.
mashir@NIS-PK564 ~ % kubectl config current-context minikube
mashir@NIS-PK564 ~ % kubectl get po -n falco
NAME          READY   STATUS    RESTARTS   AGE
falco-2bbpw   0/2     Init:0/2   0           22s
mashir@NIS-PK564 ~ % kubectl wait pods --for=condition=Ready --all -n falco
error: timed out waiting for the condition on pods/falco-2bbpw
mashir@NIS-PK564 ~ % kubectl wait pods --for=condition=Ready --all -n falco
^C
mashir@NIS-PK564 ~ % kubectl get po -n falco
NAME          READY   STATUS    RESTARTS   AGE
falco-2bbpw   0/2     Init:0/2   0           2027s
mashir@NIS-PK564 ~ % kubectl get po -n falco
NAME          READY   STATUS    RESTARTS   AGE
falco-2bbpw   0/2     Init:1/2   0           5641s
mashir@NIS-PK564 ~ % kubectl get po -n falco
NAME          READY   STATUS    RESTARTS   AGE
falco-2bbpw   2/2     Running   0           8641s
mashir@NIS-PK564 ~ % kubectl create deployment nginx --image=nginx
deployment.apps/nginx created
mashir@NIS-PK564 ~ % kubectl exec -it $(kubectl get pods --selector=app=nginx -o name) -- cat /etc/shadow
error: Internal error occurred: unable to upgrade connection: container not found ("nginx")
mashir@NIS-PK564 ~ % kubectl get po
NAME          READY   STATUS    RESTARTS   AGE
nginx-56c45fd5ff-jcn7k   0/1     ContainerCreating   0           20s
mashir@NIS-PK564 ~ % kubectl get po -w
NAME          READY   STATUS    RESTARTS   AGE
nginx-56c45fd5ff-jcn7k   0/1     ContainerCreating   0           26s
nginx-56c45fd5ff-jcn7k   1/1     Running            0           44s
^C
mashir@NIS-PK564 ~ % kubectl exec -it $(kubectl get pods --selector=app=nginx -o name) -- cat /etc/shadow
root:*:20714:0:99999:7:::
daemon:*:20714:0:99999:7:::
bin:*:20714:0:99999:7:::
sys:*:20714:0:99999:7:::
sync:*:20714:0:99999:7:::
games:*:20714:0:99999:7:::
man:*:20714:0:99999:7:::
lp:*:20714:0:99999:7:::
mail:*:20714:0:99999:7:::
news:*:20714:0:99999:7:::
uucp:*:20714:0:99999:7:::
proxy:*:20714:0:99999:7:::
www-data:*:20714:0:99999:7:::
```

Enabling falcosidekick plugin, sidekick web ui, and Triggering First Threat Event using default Rule by intentionally reading sensitive /etc/shadow file:

```
~/Documents/Learning/falco -- zsh -- zsh • kubectll1.35.1 ... -- zsh +
mashir@NIS-PK564 ~ % kubectl logs -l app=kubernetes.io/name=falco -n falco -c falco | grep Warning
08:06:31.187961023: Warning Sensitive file opened for reading by non-trusted program | file=/etc/shadow gparent=<NA> gpparent=<NA> evt_type=openat user=root user_uid=0 user_loginuid=-1 process=cat proc_exepath=/usr/bin/cat
parent=containerd-shim command=cat /etc/shadow terminal=34816 container_id=92dc093d6929 container_name=88s_nginx_nginx-56c45fd5ff-jcn7k_default_767d743-dd2d-4ae9-bb26-082313eb57a8_0 container_image_repository=nginx container_image_tag=la
test 88s_pod_name=nginx-56c45fd5ff-jcn7k 88s_ns_name=default
mashir@NIS-PK564 ~ % helm upgrade --namespace falco falco falcosecurity/falco --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true
Release "falco" has been upgraded. Happy Helming!
NAME: falco
LAST DEPLOYED: Mon Sep 21 13:07:49 2026
NAMESPACE: falco
STATUS: deployed
REVISION: 2
DESCRIPTION: Upgrade complete
NOTES:
Falco agents are spinning up on each node in your cluster. After a few
seconds, they are going to start monitoring your containers looking for
security issues.
mashir@NIS-PK564 ~ % kubectl -n falco get svc -w
NAME          TYPE        CLUSTER-IP      EXTERNAL-IP      PORT(S)          AGE
falco-falcosidekick   ClusterIP   10.104.115.53   <none>            2881/TCP,2810/TCP  33s
falco-falcosidekick-ui ClusterIP   10.111.124.39   <none>            2882/TCP         33s
falco-falcosidekick-ui-reis ClusterIP   10.106.207.114 <none>            6379/TCP         33s
^C
mashir@NIS-PK564 ~ % kubectl get po -n falco
NAME          READY   STATUS    RESTARTS   AGE
falco-7qnqh   2/2     Running   0           3m8s
falco-falcosidekick-574bffb676-bvtpc   0/1     ContainerCreating   0           3m14s
falco-falcosidekick-574bffb676-k5kfc   1/1     Running             0           3m14s
falco-falcosidekick-ui-f65b6dbbd-bdlcb 0/1     Init:0/1            0           3m14s
falco-falcosidekick-ui-f65b6dbbd-mmxwk 0/1     Init:0/1            0           3m14s
falco-falcosidekick-ui-reis-0           0/1     ContainerCreating   0           3m13s
mashir@NIS-PK564 ~ % kubectl get po -n falco
NAME          READY   STATUS    RESTARTS   AGE
falco-7qnqh   2/2     Running   0           3m37s
falco-falcosidekick-574bffb676-bvtpc   0/1     ContainerCreating   0           3m43s
falco-falcosidekick-574bffb676-k5kfc   1/1     Running             0           3m43s
falco-falcosidekick-ui-f65b6dbbd-bdlcb 0/1     Init:0/1            0           3m43s
falco-falcosidekick-ui-f65b6dbbd-mmxwk 0/1     Init:0/1            0           3m43s
falco-falcosidekick-ui-reis-0           0/1     ContainerCreating   0           3m42s
mashir@NIS-PK564 ~ % kubectl get po -n falco -w
NAME          READY   STATUS    RESTARTS   AGE
falco-7qnqh   2/2     Running   0           3m48s
falco-falcosidekick-574bffb676-bvtpc   0/1     ContainerCreating   0           3m46s
falco-falcosidekick-574bffb676-k5kfc   1/1     Running             0           3m46s
falco-falcosidekick-ui-f65b6dbbd-bdlcb 0/1     Init:0/1            0           3m46s
falco-falcosidekick-ui-f65b6dbbd-mmxwk 0/1     Init:0/1            0           3m46s
falco-falcosidekick-ui-reis-0           0/1     ContainerCreating   0           3m45s
falco-falcosidekick-ui-f65b6dbbd-bdlcb 0/1     Init:0/1            0           5n31s
falco-falcosidekick-574bffb676-bvtpc   0/1     Running             0           5n32s
falco-falcosidekick-ui-f65b6dbbd-mmxwk 0/1     Init:0/1            0           5n34s
falco-falcosidekick-ui-reis-0           0/1     Running             0           5n35s
falco-falcosidekick-ui-reis-0           1/1     Running             0           5n41s
falco-falcosidekick-574bffb676-bvtpc   0/1     Running             0           5n43s
falco-falcosidekick-ui-f65b6dbbd-bdlcb 0/1     PodInitializing     0           5n47s
falco-falcosidekick-ui-f65b6dbbd-mmxwk 0/1     PodInitializing     0           5n47s
falco-falcosidekick-ui-f65b6dbbd-bdlcb 0/1     Running              0           6n23s
falco-falcosidekick-ui-f65b6dbbd-mmxwk 0/1     Running              0           6n23s
falco-falcosidekick-ui-f65b6dbbd-bdlcb 1/1     Running              0           6n32s
falco-falcosidekick-ui-f65b6dbbd-mmxwk 1/1     Running              0           6n34s
^C
mashir@NIS-PK564 ~ % kubectl get po -n falco -w
NAME          READY   STATUS    RESTARTS   AGE
falco-7qnqh   2/2     Running   0           8m43s
falco-falcosidekick-574bffb676-bvtpc   1/1     Running             0           8m49s
falco-falcosidekick-574bffb676-k5kfc   1/1     Running             0           8m49s
falco-falcosidekick-ui-f65b6dbbd-bdlcb 1/1     Running             0           8m49s
falco-falcosidekick-ui-f65b6dbbd-mmxwk 1/1     Running             0           8m49s
falco-falcosidekick-ui-reis-0           1/1     Running             0           8m48s
^C
mashir@NIS-PK564 ~ % kubectl exec -it $(kubectl get pods --selector=app=nginx -o name) -- cat /etc/shadow
root:*:20714:0:99999:7:::
daemon:*:20714:0:99999:7:::
```

Port-forwarding the falcosidekick web ui service to observe the event in UI (we can also observe the event in pod container logs):

[illegible]

The falcosidekick web ui is placed in the bottom with all the Events produced in the Exercise along with this Event:

Upgrading the falco helm release with writing custom rule to detect file creation under /etc/ directory:

mail*:*:20714:0:99999:7:::	~/Documents/Learning/falco -- -zsh	~ -- -zsh • kubectlf1.35.1	~ -- -zsh	+
news*:*:20714:0:99999:7:::				
uucp*:*:20714:0:99999:7:::				
proxy*:*:20714:0:99999:7:::				
www-data*:*:20714:0:99999:7:::				
Backup*:*:20714:0:99999:7:::				
list*:*:20714:0:99999:7:::				
irc*:*:20714:0:99999:7:::				
_apt*:*:20714:0:99999:7:::				
ngbody*:*:20714:0:99999:7:::				
nginx*:*:20715:1:::				
meshir@NIS-PK564 ~ % ls	Desktop	Documents	Downloads	IdeaProjects
meshir@NIS-PK564 ~ % cd Documents				Library
meshir@NIS-PK564 Documents % ls				Movies
Colina_Installation_01.md				Music
Colina_Installation_02.md				Pictures
DevOps Internship Projects				Postman
DevOps Internship Projects - Nisum OSI.zip				Public
meshir@NIS-PK564 Documents % cd Learning				VirtualBox VMs
meshir@NIS-PK564 Learning % ls				
ArgoCD HashiCorpVault Integration				
Azure-Container-Apps				
JSAC				
meshir@NIS-PK564 Learning % mkdir falco				
meshir@NIS-PK564 Learning % cd falco				
meshir@NIS-PK564 falco % nano falco_custom_rules_cm.yaml				
meshir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml				
Error: failed to parse falco_custom_rules_cm.yaml: cannot unmarshal yaml document: error converting YAML to JSON: yaml: line 10: could not find expected ':'				
meshir@NIS-PK564 falco % ls				
meshir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true				
Error: failed to parse falco_custom_rules_cm.yaml: cannot unmarshal yaml document: error converting YAML to JSON: yaml: line 10: could not find expected ':'				
meshir@NIS-PK564 falco % nano falco_custom_rules_cm.yaml				
meshir@NIS-PK564 falco % nano falco_custom_rules_cm.yaml				
meshir@NIS-PK564 falco % nano falco_custom_rules_cm.yaml				
meshir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true				
Release "falco" has been upgraded. Happy Helming!				
NAME: falco				
LAST DEPLOYED: Mon Sep 21 13:34:27 2026				
NAMESPACE: falco				
STATUS: deployed				
REVISION: 3				
DESCRIPTION: Upgrade complete				
NOTES:				
Falco agents are spinning up on each node in your cluster. After a few seconds, they are going to start monitoring your containers looking for security issues.				
meshir@NIS-PK564 falco % helm lint falcosecurity/falco -f falco_custom_rules_cm.yaml				
=> Linting falcosecurity/falco				
Error unable to check Chart.yaml file in chart: stat falcosecurity/falco/Chart.yaml: no such file or directory				
Error: 1 chart(s) linted, 1 chart(s) failed				
meshir@NIS-PK564 falco % helm lint falcosecurity/falco -f falco_custom_rules_cm.yaml				
=> Linting falcosecurity/falco				
Error unable to check Chart.yaml file in chart: stat falcosecurity/falco/Chart.yaml: no such file or directory				
Error: 1 chart(s) linted, 1 chart(s) failed				
meshir@NIS-PK564 falco % kubectf exec -it \$(kubectf get pods --selector=app=nginx -o name) -- touch /etc/test_file_for_falco_rule				
meshir@NIS-PK564 falco % ls				
falco_custom_rules_cm.yaml				
meshir@NIS-PK564 falco % nano falco_email-values.yaml				
meshir@NIS-PK564 falco %				
meshir@NIS-PK564 falco %				
meshir@NIS-PK564 falco % ls				
falco-email-values.yaml				
meshir@NIS-PK564 falco % mv falco_email-values.yaml falco_email-values.yaml				
meshir@NIS-PK564 falco % ls				
falco_email-values.yaml				
meshir@NIS-PK564 falco %				

The rule helm value file is present at the project directory at:
/ falco_custom_rules_cm.yaml

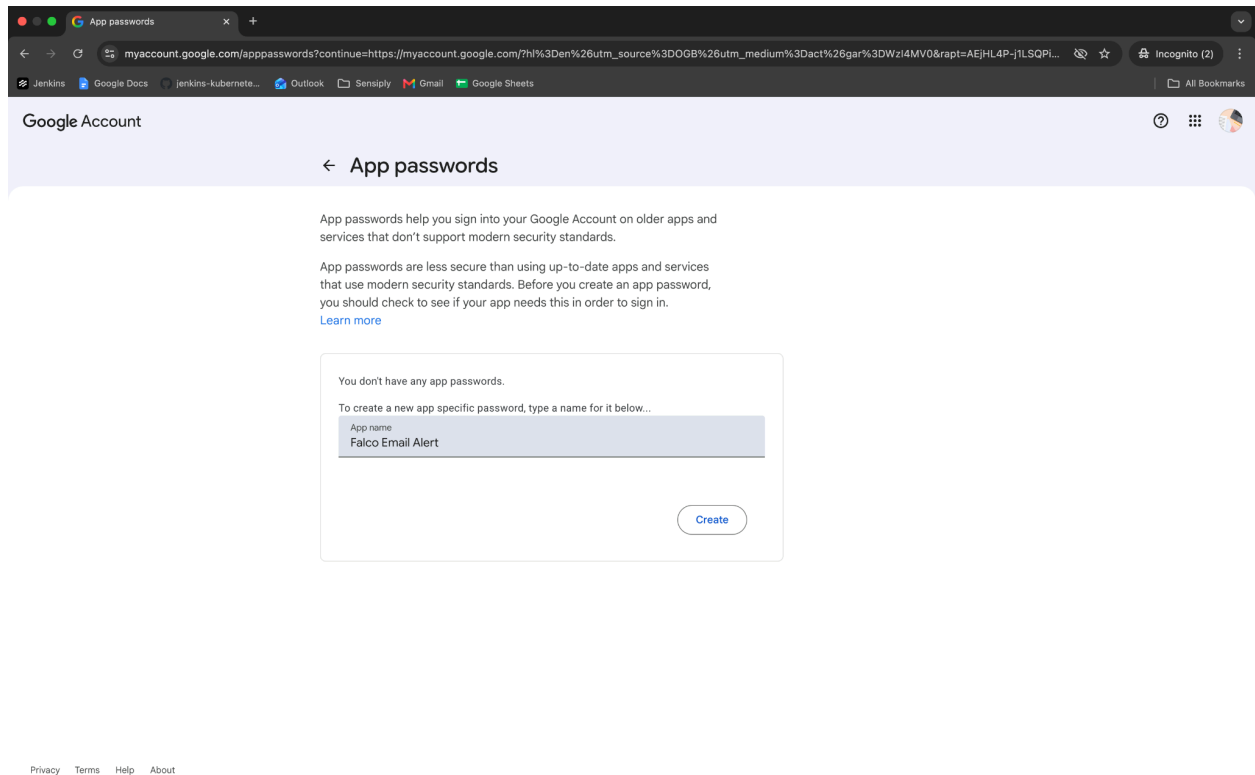
The falcosidekick web ui is placed in the bottom with all the Events produced in the Exercise along with this Event:

Configuring Email Notifications on event trigger by writing and passing helm values file containing email configuration and upgrading the falco helm release:

~/Documents/Learning/falco -- -zsh	~/Documents/Learning/falco -- nano falco_custom_rules_cm.yaml	-- -zsh - kubectll3.5.1	~/Documents/Learning/falco -- -zsh	+
Error: 1 chart(s) linted, 1 chart(s) failed				
mashir@NIS-PK564 falco % helm lint falcosecurity/falco -f falco_custom_rules_cm.yaml				
=> Linting falcosecurity/falco				
Error: unable to check Chart.yaml file in chart: stat falcosecurity/falco/Chart.yaml: no such file or directory				
Error: 1 chart(s) linted, 1 chart(s) failed				
mashir@NIS-PK564 falco % kubectll exec -it \$(kubectll get pods --selector=app=nginx -o name) -- touch /etc/test_file_for_falco_rule				
mashir@NIS-PK564 falco % ls				
falco_custom_rules_cm.yaml				
mashir@NIS-PK564 falco % nano falco_email-values.yaml				
mashir@NIS-PK564 falco %				
mashir@NIS-PK564 falco %				
mashir@NIS-PK564 falco % ls				
falco_email-values.yaml				
mashir@NIS-PK564 falco % mv falco_email-values.yaml falco_email-values.yaml				
mashir@NIS-PK564 falco % ls				
falco_email-values.yaml				
mashir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml -f falco_email-values.yaml --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true				
Release "falco" has been upgraded. Happy Helming!				
NAME: falco				
LAST DEPLOYED: Mon Sep 21 15:21:15 2026				
NAMESPACE: falco				
STATUS: deployed				
REVISION: 4				
DESCRIPTION: Upgrade complete				
NOTES:				
Falco agents are spinning up on each node in your cluster. After a few seconds, they are going to start monitoring your containers looking for security issues.				
mashir@NIS-PK564 falco % kubectll exec -it \$(kubectll get pods --selector=app=nginx -o name) -- cat /etc/shadow				
root:*:20714:0:99999:7:::				
daemon:*:20714:0:99999:7:::				
bin:*:20714:0:99999:7:::				
sys:*:20714:0:99999:7:::				
sync:*:20714:0:99999:7:::				
games:*:20714:0:99999:7:::				
man:*:20714:0:99999:7:::				
lp:*:20714:0:99999:7:::				
mail:*:20714:0:99999:7:::				
news:*:20714:0:99999:7:::				
uucp:*:20714:0:99999:7:::				
proxy:*:20714:0:99999:7:::				
www-data:*:20714:0:99999:7:::				
backup:*:20714:0:99999:7:::				
list:*:20714:0:99999:7:::				
irc:*:20714:0:99999:7:::				
_apt:*:20714:0:99999:7:::				
nobody:*:20714:0:99999:7:::				
nginx:*:20715:1:::				
mashir@NIS-PK564 falco % nano falco_custom_rules_cm-2-gcp-creds.yaml				
mashir@NIS-PK564 falco % ls				
falco_email-values.yaml				
falco_custom_rules_cm-2-gcp-creds.yaml				
falco_custom_rules_cm.yaml				
mashir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml -f falco_email-values.yaml -f falco_custom_rules_cm-2-gcp-creds.yaml --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true				
i.enabled=true				
Error: failed to parse falco_custom_rules_cm-2-gcp-creds.yaml: cannot unmarshal yaml document: error unmarshaling JSON: while decoding JSON: json: cannot unmarshal array into Go value of type map[string]Interface {}				
mashir@NIS-PK564 falco % nano falco_custom_rules_cm-2-gcp-creds.yaml				
mashir@NIS-PK564 falco % nano falco_custom_rules_cm-2-gcp-creds.yaml				
mashir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml -f falco_email-values.yaml -f falco_custom_rules_cm-2-gcp-creds.yaml --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true				
i.enabled=true				
Release "falco" has been upgraded. Happy Helming!				
NAME: falco				
LAST DEPLOYED: Mon Sep 21 17:28:29 2026				
NAMESPACE: falco				
STATUS: deployed				
REVISION: 5				
DESCRIPTION: Upgrade complete				
NOTES:				
Falco agents are spinning up on each node in your cluster. After a few seconds, they are going to start monitoring your containers looking for security issues.				
mashir@NIS-PK564 falco % █				

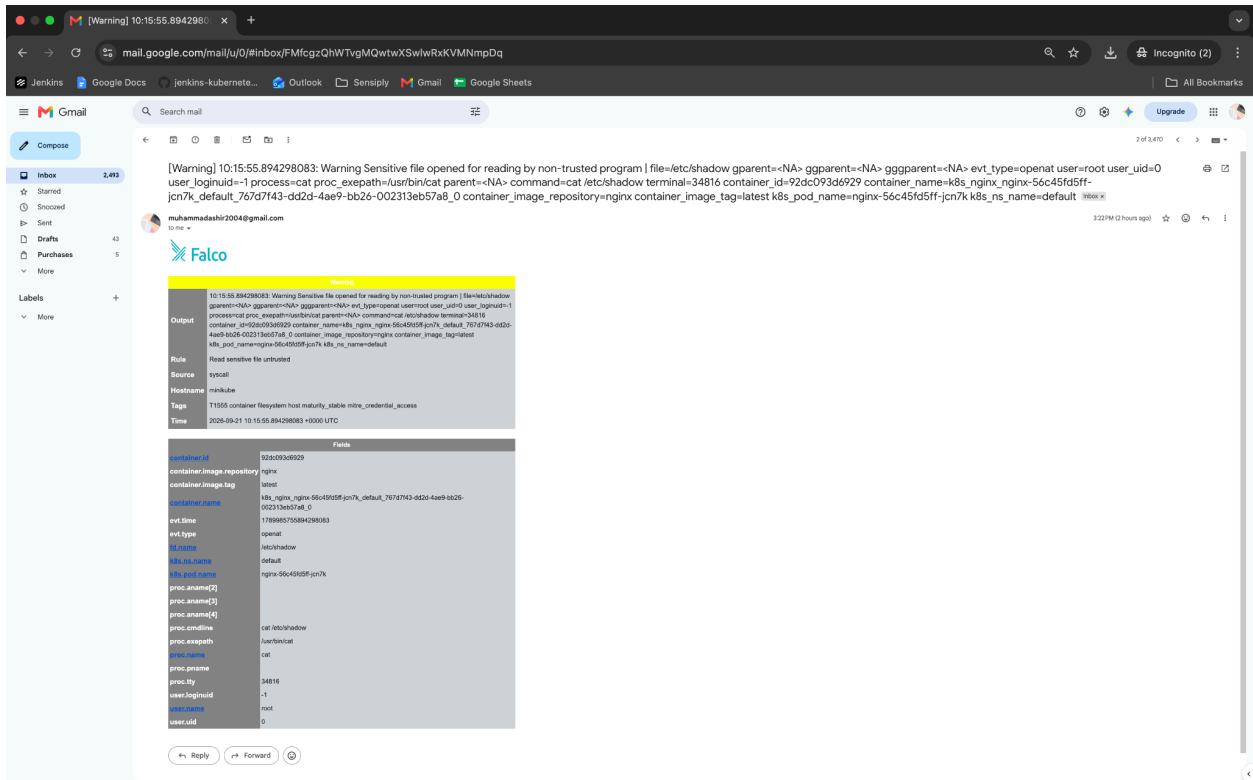
The helm value file for email configuration is at the project directory at:
/ falco-email-values.yaml

Generated Google App Password for gmail:



The falcosidekick web ui is placed in the bottom with all the Events produced in the Exercise along with this Event:

Email Alert Notification Received for the Event Detected:



Configuring a Custom Rule to detect processes searching for GCP credential activity:

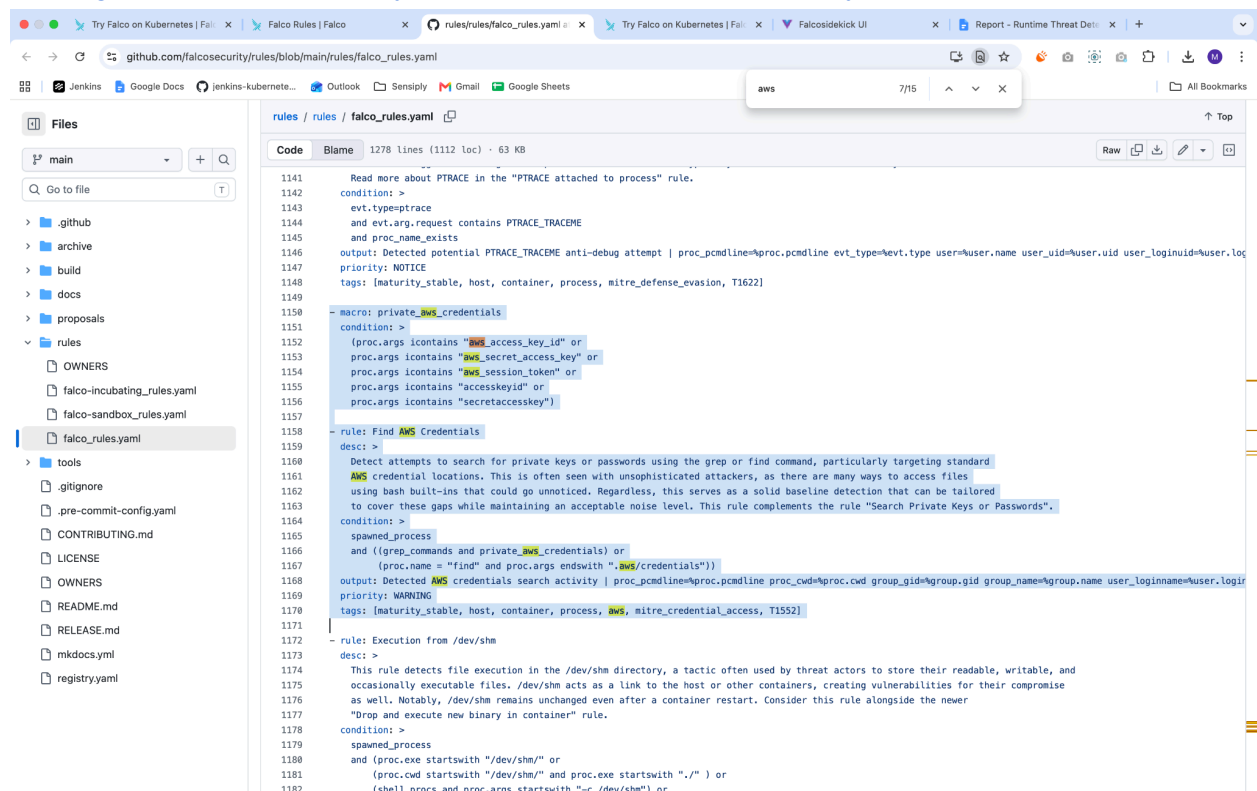
```
~/Documents/Learning/falco -- -zsh      ~/Documents/Learning/falco -- nano falco_custom_rules_cm.yaml      ~ -- -zsh - kubect1.35.1      ~/Documents/Learning/falco -- -zsh +
Error: 1 chart(s) linted, 1 chart(s) failed
mashir@NIS-PK564 falco % helm lint falcosecurity/falco -f falco_custom_rules_cm.yaml
=> Linting falcosecurity/falco
Error: unable to check Chart.yaml file in chart: stat falcosecurity/falco/Chart.yaml: no such file or directory

Error: 1 chart(s) linted, 1 chart(s) failed
mashir@NIS-PK564 falco % kubect1 exec -it $(kubect1 get pods --selector=app=nginx -o name) -- touch /etc/test_file_for_falco_rule

mashir@NIS-PK564 falco % ls
falco_custom_rules_cm.yaml
mashir@NIS-PK564 falco % nano falco_email_values.yaml
mashir@NIS-PK564 falco %
mashir@NIS-PK564 falco %
mashir@NIS-PK564 falco %
falco_email_values.yaml      falco_custom_rules_cm.yaml
mashir@NIS-PK564 falco % mv falco_email_values.yaml falco_email_values.yaml
mashir@NIS-PK564 falco % ls
falco_email_values.yaml      falco_custom_rules_cm.yaml
mashir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml -f falco_email_values.yaml --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true
Release 'falco' has been upgraded. Happy Helming!
NAME: falco
LAST DEPLOYED: Mon Sep 21 15:21:15 2026
NAMESPACE: falco
STATUS: deployed
REVISION: 4
DESCRIPTION: Upgrade complete
NOTES:
Falco agents are spinning up on each node in your cluster. After a few
seconds, they are going to start monitoring your containers looking for
security issues.
mashir@NIS-PK564 falco % kubect1 exec -it $(kubect1 get pods --selector=app=nginx -o name) -- cat /etc/shadow
root:*:20714:0:99999:7:::
daemon:*:20714:0:99999:7:::
bin:*:20714:0:99999:7:::
sys:*:20714:0:99999:7:::
sync:*:20714:0:99999:7:::
games:*:20714:0:99999:7:::
man:*:20714:0:99999:7:::
lp:*:20714:0:99999:7:::
mail:*:20714:0:99999:7:::
news:*:20714:0:99999:7:::
uucp:*:20714:0:99999:7:::
proxy:*:20714:0:99999:7:::
www-data:*:20714:0:99999:7:::
backup:*:20714:0:99999:7:::
list:*:20714:0:99999:7:::
irc:*:20714:0:99999:7:::
_apt:*:20714:0:99999:7:::
nobody:*:20714:0:99999:7:::
nginx:::20715:::
mashir@NIS-PK564 falco % nano falco_custom_rules_cm-2-gcp-creds.yaml
mashir@NIS-PK564 falco % ls
falco_email_values.yaml      falco_custom_rules_cm.yaml
falco_email_values.yaml      falco_custom_rules_cm.yaml
mashir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml -f falco_email_values.yaml -f falco_custom_rules_cm-2-gcp-creds.yaml --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true
Error: failed to parse falco_custom_rules_cm-2-gcp-creds.yaml: cannot unmarshal yaml document: error unmarshaling JSON: while decoding JSON: json: cannot unmarshal array into Go value of type map[string]interface {}
mashir@NIS-PK564 falco % nano falco_custom_rules_cm-2-gcp-creds.yaml
mashir@NIS-PK564 falco % nano falco_custom_rules_cm-2-gcp-creds.yaml
mashir@NIS-PK564 falco % helm upgrade --namespace falco falco falcosecurity/falco -f falco_custom_rules_cm.yaml -f falco_email_values.yaml -f falco_custom_rules_cm-2-gcp-creds.yaml --set falcosidekick.enabled=true --set falcosidekick.webui.enabled=true
Release 'falco' has been upgraded. Happy Helming!
NAME: falco
LAST DEPLOYED: Mon Sep 21 17:28:29 2026
NAMESPACE: falco
STATUS: deployed
REVISION: 5
DESCRIPTION: Upgrade complete
NOTES:
Falco agents are spinning up on each node in your cluster. After a few
seconds, they are going to start monitoring your containers looking for
security issues.
mashir@NIS-PK564 falco %
```

For this rule I actually referred to official default rule defined by Falco which is actually for AWS Credentials at:

https://github.com/falcosecurity/rules/blob/main/rules/falco_rules.yaml



```
1141 Read more about PTRACE in the "PTRACE attached to process" rule.
1142 condition: >
1143   evt.type=ptrace
1144   and evt.arg.request contains PTRACE_TRACEME
1145   and proc_name_exists
1146 output: Detected potential PTRACE_TRACEME anti-debug attempt | proc_pcmdline=proc.pcmdline evt_type=evt.type user=user.name user_uid=user.uid user_loginuid=user.log
1147 priority: NOTICE
1148 tags: [maturity_stable, host, container, process, mitre_defense_evasion, T1622]
1149
1150 - macro: private_aws_credentials
1151 condition: >
1152   (proc.args icontains "aws_access_key_id" or
1153    proc.args icontains "aws_secret_access_key" or
1154    proc.args icontains "aws_session_token" or
1155    proc.args icontains "accessKeyId" or
1156    proc.args icontains "secretaccesskey")
1157
1158 - rule: Find AWS Credentials
1159 desc: >
1160   Detect attempts to search for private keys or passwords using the grep or find command, particularly targeting standard
1161   AWS credential locations. This is often seen with unsophisticated attackers, as there are many ways to access files
1162   using bash built-ins that could go unnoticed. Regardless, this serves as a solid baseline detection that can be tailored
1163   to cover these gaps while maintaining an acceptable noise level. This rule complements the rule "Search Private Keys or Passwords".
1164 condition: >
1165   spawned_process
1166   and ((grep_commands and private_aws_credentials) or
1167        (proc.name = "find" and proc.args endswith ".aws/credentials"))
1168 output: Detected AWS credentials search activity | proc_pcmdline=proc.pcmdline proc_cwd=proc.cwd group_gid=group.gid group_name=group.name user_loginname=user.logi
1169 priority: WARNING
1170 tags: [maturity_stable, host, container, process, aws, mitre_credential_access, T1552]
1171
1172 - rule: Execution from /dev/shm
1173 desc: >
1174   This rule detects file execution in the /dev/shm directory, a tactic often used by threat actors to store their readable, writable, and
1175   occasionally executable files. /dev/shm acts as a link to the host or other containers, creating vulnerabilities for their compromise
1176   as well. Notably, /dev/shm remains unchanged even after a container restart. Consider this rule alongside the newer
1177   "Drop and execute new binary in container" rule.
1178 condition: >
1179   spawned_process
1180   and (proc.exe startswith "/dev/shm/" or
1181        (proc.cwd startswith "/dev/shm/" and proc.exe startswith "./" ) or
1182        (shell nrpc and proc.args startswith "-c /dev/shm/")) or
```

I created the custom rule to detect if any process attempts to search for GCP Credentials using standard GCP Credentials environment variables & file location. First defined macro containing common gcp credentials values, then defined a rule to detect if any process including grep or find process searching for the values

My helm value file for detecting GCP Credentials discovery activity is at the project directory at:

/ falco_custom_rules_cm-2-gcp-creds.yaml

The falcosidekick web ui is placed in the bottom with all the Events produced in the Exercise along with this Event

Email Alert Notification Received for the Event Detected:

[Warning] 12:32:51.227642871: Warning Detected GCP credentials search activity | proc_pcmdline=<NA> proc_cwd=/ group_gid=0 group_name=root user_loginname=<NA> evt_type=execve user=root user_uid=0 user_loginuid=-1 container_id=92dc093d6929 container_name=k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 container_image_repository=nginx container_image_tag=latest k8s_pod_name=nginx-56c45fd5ff-jcn7k k8s_ns_name=default

muhammadashir2004@gmail.com
to me

Warning

12:32:51.227642871: Warning Detected GCP credentials search activity | proc_pcmdline=<NA> proc_cwd=/ group_gid=0 group_name=root user_loginname=<NA> evt_type=execve user=root user_uid=0 user_loginuid=-1 container_id=92dc093d6929 container_name=k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 container_image_repository=nginx container_image_tag=latest k8s_pod_name=nginx-56c45fd5ff-jcn7k k8s_ns_name=default

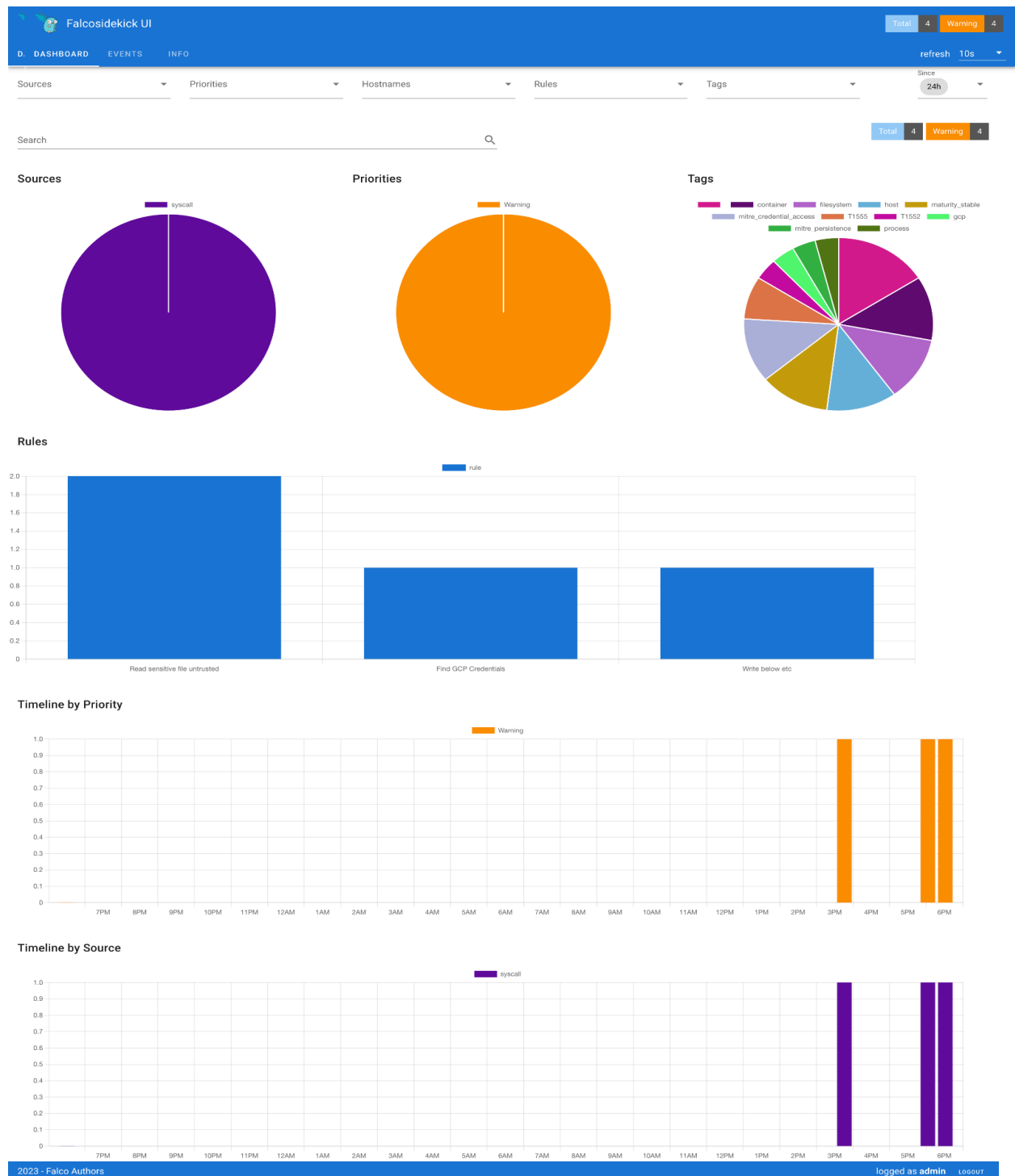
Rule Find GCP Credentials
Source syscalls
Hostname minikube
Tags T1552 container gcp host maturity_stable mitre_credential_access process
Time 2026-09-21 12:32:51.227642871 +0000 UTC

Fields

container.id	92dc093d6929
container.image.repository	nginx
container.image.tag	latest
container.name	k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0
evt.time	1789993971227642871
evt.type	execve
group.gid	0
group.name	root
k8s.ns.name	default
k8s.pod.name	nginx-56c45fd5ff-jcn7k
proc.cwd	/
proc.pcmdline	<NA>
user.loginname	<NA>
user.loginuid	-1
user.name	root
user.uid	0

Reply Forward

Dashboard:



Events Detected:

Falcosidekick UI

D. DASHBOARD

EVENTS

INFO

Total

4

Warning

4

refresh 10s

Sources

Priorities

Hostnames

Rules

Tags

Since 24h

Search

Total 4 Warning 4

EXPORT

Timestamp	Source	Hostname	Priority	Rule	Output	Tags
2026/09/21 17:32:51.227	syscall	minikube	Warning	Find GCP Credentials	12:32:51.227642871: Warning Detected GCP credentials search activity proc.pcmldine=<NA> proc.cwd=/ group_gid=0 group_name=root user_loginname=<NA> evl_type=execve user=root user_uid=0 user_loginuid=1 container_id=92dc093d6929 container_name=k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 container_image_repository=nginx container_image_tag=latest k8s_pod_name=nginx-56c45fd5ff-jcn7k k8s_ns_name=default container_id 92dc093d6929 container.image.repository nginx container.image.tag latest container.name k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 evl.time 1789993971227643000 evl.type execve group_gid 0 group.name root k8s.ns.name default k8s.pod.name nginx-56c45fd5ff-jcn7k proc.cwd / proc.pcmldine user.loginname <NA> user.loginuid -1 user.name root user.uid 0	T1552 container gcp host maturity_stable mitre_credential_access process
2026/09/21 15:15:55.894	syscall	minikube	Warning	Read sensitive file untrusted	10:15:55.894298083: Warning Sensitive file opened for reading by non-trusted program file=/etc/shadow gparent=<NA> ggparent=<NA> ggpparent=<NA> evl_type=opentat user=root user_uid=0 user_loginuid=1 process=cat proc_exepath=/usr/bin/cat parent=<NA> command=cat /etc/shadow terminal=34816 container_id=92dc093d6929 container_name=k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 container_image_repository=nginx container_image_tag=latest k8s_pod_name=nginx-56c45fd5ff-jcn7k k8s_ns_name=default container_id 92dc093d6929 container.image.repository nginx container.image.tag latest container.name k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 evl.time 1789985755894298000 evl.type opentat fd name /etc/shadow k8s.ns.name default k8s.pod.name nginx-56c45fd5ff-jcn7k proc.name[2] proc.name[3] proc.name[4] proc.cmdline cat /etc/shadow proc.exepath /usr/bin/cat proc.name cat proc.pname proc.tty 34816 user.loginuid -1 user.name root user.uid 0	T1555 container filesystem host maturity_stable mitre_credential_access
2026/09/21 13:36:20.991	syscall	minikube	Warning	Write below etc	08:36:20.991555128: Warning File below /etc opened for writing file=/etc/test_file_for_falco_rule pcmldine=<NA> gparent=<NA> ggparent=<NA> ggpparent=<NA> evl_type=opentat user=root user_uid=0 user_loginuid=1 process=touch proc_exepath=/usr/bin/touch parent=<NA> command=touch /etc/test_file_for_falco_rule terminal=34816 container_id=92dc093d6929 container_name=k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 container_image_repository=nginx container_image_tag=latest k8s_pod_name=nginx-56c45fd5ff-jcn7k k8s_ns_name=default container_id 92dc093d6929 container.image.repository nginx container.image.tag latest container.name k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 evl.time 1789979780991555000 evl.type opentat fd name /etc/test_file_for_falco_rule k8s.ns.name default k8s.pod.name nginx-56c45fd5ff-jcn7k proc.name[2] proc.name[3] proc.name[4] proc.cmdline touch /etc/test_file_for_falco_rule proc.exepath /usr/bin/touch proc.name touch proc.pcmldine proc.pname proc.tty 34816 user.loginuid -1 user.name root user.uid 0	filesystem mitre_persistence
2026/09/21 13:20:03.922	syscall	minikube	Warning	Read sensitive file untrusted	08:20:03.922169971: Warning Sensitive file opened for reading by non-trusted program file=/etc/shadow gparent=<NA> ggparent=<NA> ggpparent=<NA> evl_type=opentat user=root user_uid=0 user_loginuid=1 process=cat proc_exepath=/usr/bin/cat parent=<NA> command=cat /etc/shadow terminal=34816 container_id=92dc093d6929 container_name=k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 container_image_repository=nginx container_image_tag=latest k8s_pod_name=nginx-56c45fd5ff-jcn7k k8s_ns_name=default container_id 92dc093d6929 container.image.repository nginx container.image.tag latest container.name k8s_nginx_nginx-56c45fd5ff-jcn7k_default_767d7f43-dd2d-4ae9-bb26-002313eb57a8_0 evl.time 1789978803922169900 evl.type opentat fd name /etc/shadow k8s.ns.name default k8s.pod.name nginx-56c45fd5ff-jcn7k proc.name[2] proc.name[3] proc.name[4] proc.cmdline cat /etc/shadow proc.exepath /usr/bin/cat proc.name cat proc.pname proc.tty 34816 user.loginuid -1 user.name root user.uid 0	T1555 container filesystem host maturity_stable mitre_credential_access

Rows per page: 10

1 4 of 4

logged as admin Logout

I exported the Events json file locally, and is present at project root directory at:

/ falco_events.json